

ชื่อเรื่อง: ปัจจัยที่มีผลต่อการเสียชีวิตของผู้ติดเชื้อเอชไอวีที่เริ่มยาต้านไวรัสในประเทศไทย โดยใช้โมเดลจำลองแบบ Stratified Cox Proportional Hazards Factor Association of Mortality Risk Among People Living with HIV Antiretroviral Therapy Initiation in Thailand: A Stratified Cox Proportional Hazards Model

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ข้อมูลที่ใช้ศึกษา:	
Cross sectional data (ข้อมูลภาคตัดขวาง) ปี 2021 - 2022	
การแบ่งกลุ่มตัวแปรอิสระที่สนใจ:	
sex แบ่งเป็น 2 กลุ่มคือ 0 แทน Male และ 1 แทน Female firstoi_g แบ่งกลุ่ม 3 กลุ่มคือ 0 แทน Yes, 1 แทน No และ 2 แทน Unknown hivstage แบ่งกลุ่ม 4 กลุ่มคือ 0 แทน Asymptomatic HIV, 1 แทน Symptomatic HIV, 2 แทน AIDS agerartg แบ่งกลุ่ม 3 กลุ่มคือ 0 แทน 15 to 24 years, 1 แทน 25 to 49 years และ 2 แทน ≥ 50 years fartg_n แบ่งกลุ่ม 4 กลุ่มคือ 0 แทน DTG-based ART, 1 แทน NNTRI-based ART, 2 แทน PIs-based ART และ 3 แทน Others yearart_n แบ่งเป็น 2 กลุ่มคือ 0 แทน 2021 และ 1 แทน 2022 cd4blg แบ่งกลุ่ม 4 กลุ่มคือ 0 แทน < 200 (cells/mm ³), 1 แทน 200 to < 350 (cells/mm ³), 2 แทน ≥ 350 (cells/mm ³) และ 3 แทน Unknown region_g แบ่งกลุ่ม 3 กลุ่มคือ 0 แทน Bangkok, 1 แทน Central, 2 แทน Eastern, 3 แทน Northeastern, 4 แทน Northern, 5 แทน Southern du_art_g แบ่งกลุ่ม 3 กลุ่มคือ 0 แทน Within 6 months, 1 แทน > 6 to 12 months และ 2 แทน > 12 months rapid_artg แบ่งกลุ่ม 3 กลุ่มคือ 0 แทน Within 1 month, 1 แทน > 1 to 3 months และ 2 แทน > 3 months minc_group แบ่งกลุ่ม 4 กลุ่มคือ 0 แทน USS, 1 แทน SSS, 2 แทน WEL&OFC และ 3 แทน Others	
ตัวแปรตามที่สนใจ: ตัวแปรเชิงคุณภาพ	
died - สถานะของผู้ป่วยขณะรับการรักษา แบ่งเป็น 2 กลุ่มคือ 0 แทน Alive และ 1 แทน Dead	
ตัวแปรอิสระที่สนใจ:	
sex - เพศ (ตัวแปรเชิงคุณภาพ) firstoi_g – การติดเชื้อโรคฉวยโอกาสในระหว่างการให้ยาครั้งแรก (ตัวแปรเชิงคุณภาพ) hivstage - ระดับความรุนแรงของเชื้อ HIV (ตัวแปรเชิงคุณภาพ) agerartg - กลุ่มอายุของผู้ป่วยตอน ณ ที่เริ่มการรักษาครั้งแรกด้วยยาต้านไวรัสเอชไอวี (ตัวแปรเชิงคุณภาพ) fartg_n – กลุ่มยาที่รับในการรักษาครั้งแรก (ตัวแปรเชิงคุณภาพ) yearart_n – ปีที่เข้ารับการรักษาครั้งแรก (ตัวแปรเชิงคุณภาพ) cd4blg - ค่า CD4 เมื่อเริ่มทานยารักษาครั้งแรก (ตัวแปรเชิงคุณภาพ) region_g - เขตพื้นที่ในประเทศไทย (ตัวแปรเชิงคุณภาพ) du_art_g - กลุ่มของระยะเวลาการรักษาตั้งแต่วันที่เริ่มยารักษาจนถึงวันสุดท้าย (ตัวแปรเชิงคุณภาพ) rapid_artg - กลุ่มของผู้ป่วยที่เข้ารับการรักษาหลังจากเข้ามาในระบบ (ตัวแปรเชิงคุณภาพ) minc_group – ประเภทสิทธิประกันสุขภาพหลักของผู้ป่วย (ตัวแปรเชิงคุณภาพ)	
สถิติที่ใช้ในการศึกษา:	
Descriptive statistics	- Frequency - Percent - Mean - SD - Median - IQR - person-years - Kaplan-Meier survival curve - Goodness-of-fits - ROC curve
Inferential statistics	- Pearson's Chi-square tests - Mann-Whitney U tests - Log-rank tests - Cox regression - PH tests - Stratified Cox regression - 95%CI

Figure 1. diagram of study population.

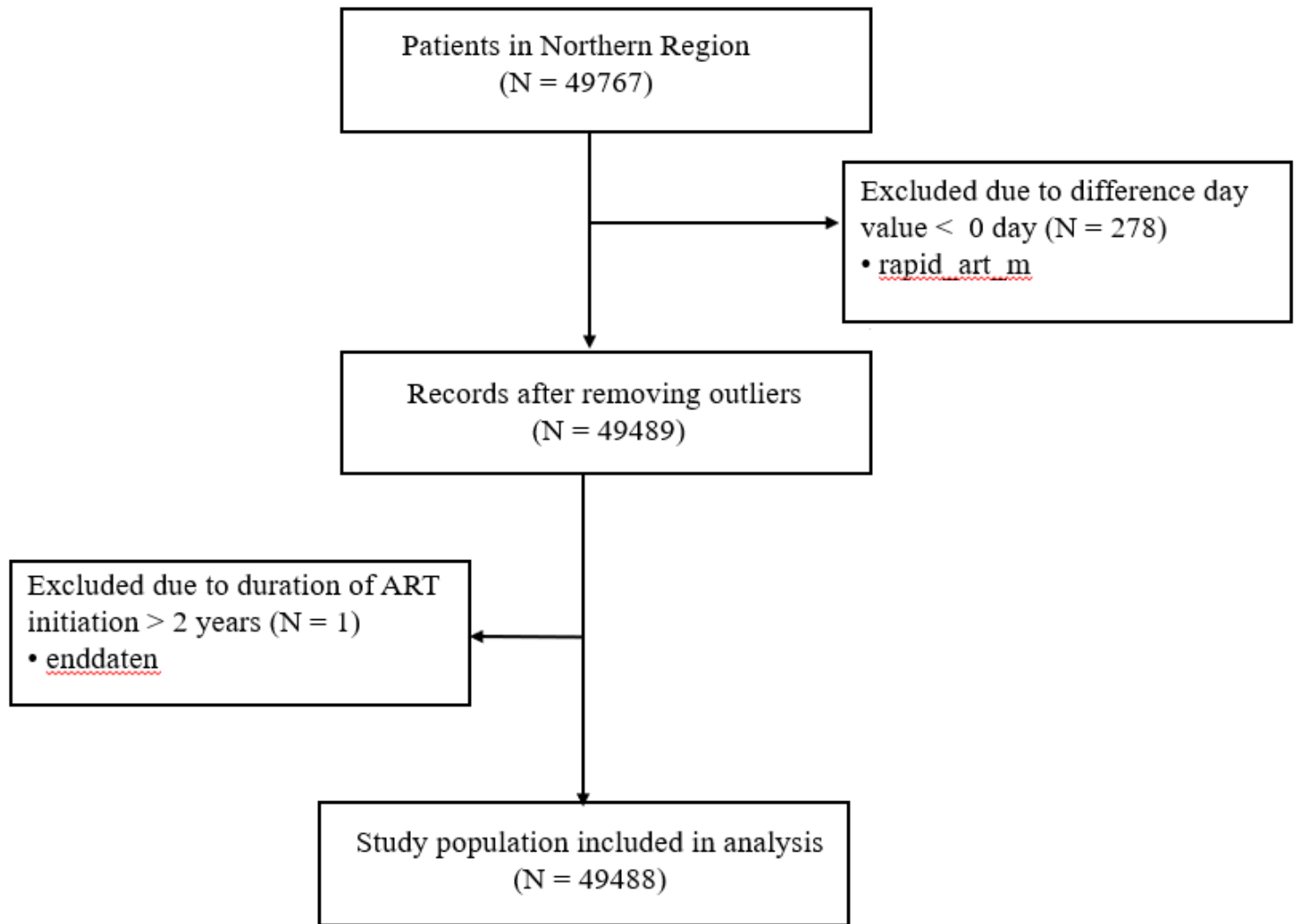


Table 1 Characteristics of people living with HIV at ART initiation in Thailand by mortality status.

Characteristics	Mortality status			<i>p</i>
	Survival	Death	Total	
Overall, N (%)	46954 (94.90)	2534 (5.10)	49488 (100)	
Gender, N (%)				0.460
Male	33819 (72.03)	1808 (71.35)	35627 (71.99)	
Female	13135 (27.97)	726 (28.65)	13861 (28.01)	
Median (IQR) age (years)	33.00 (26.00–43.00)	42.00 (32.00–50.00)	34.00 (26.00–43.00)	< 0.001*
Age group, N (%)				< 0.001*
15 to 24 years	9785 (20.84)	161 (6.35)	9946 (20.10)	
25 to 49 years	31093 (66.22)	1685 (66.50)	32778 (66.23)	
≥ 50 years	6076 (12.94)	688 (27.15)	6764 (13.67)	
Median (IQR) Pre-ART CD4 count (cells/mm³)	245.00 (78.00–435.00)	46.50 (17.00–128.00)	233.00 (68.00–425.00)	< 0.001*
Pre-ART CD4 count, N (%)				< 0.001*
< 200	17993 (38.32)	1796 (70.88)	19789 (39.99)	
200 to < 350	8956 (19.07)	190 (7.50)	9146 (18.48)	
≥ 350	14668 (31.24)	124 (4.89)	14792 (29.89)	
Unknown	5337 (11.37)	424 (16.73)	5761 (11.64)	
Median (IQR), Log₁₀ viral load after 1 year of ART (copies/ml)	1.60 (1.30–1.82)	2.14 (1.60–5.10)	1.60 (1.30–1.84)	< 0.001*
Viral load after 1 year of ART, N (%)				< 0.001*
≤ 50	22989 (48.96)	239 (9.43)	23228 (46.94)	
> 50	8784 (18.71)	309 (12.19)	9093 (18.37)	
Unknown	15181 (32.33)	1986 (78.37)	17167 (34.96)	
First regimen, N (%)				< 0.001*
NNRTI-based ART	29540 (62.91)	1911 (75.41)	31451 (63.55)	
PI-based ART	949 (2.02)	66 (2.60)	1015 (2.05)	
DTG-based ART	14498 (30.88)	453 (17.88)	14951 (30.21)	
Others	1967 (4.19)	104 (4.10)	2071 (4.18)	
Opportunistic infection at ART initiation, N (%)				< 0.001*
Yes	2893 (6.16)	434 (17.13)	3327 (6.72)	
No	44061 (93.84)	2100 (82.87)	46161 (93.28)	
Year of ART initiation, N (%)				< 0.001*
2021	24046 (51.21)	1614 (63.69)	25660 (51.85)	
2022	22908 (48.79)	920 (36.31)	23828 (48.15)	
HIV stage, N (%)				< 0.001*
Asymptomatic HIV	29394 (62.60)	1165 (45.97)	30559 (61.75)	
Symptomatic HIV	6776 (14.43)	496 (19.57)	7272 (14.69)	
AIDS	10784 (22.97)	873 (34.45)	11657 (23.56)	
Median (IQR) duration of ART initiation (months)	9.30 (3.61–16.33)	1.12 (0.03–4.14)	8.76 (3.06–15.90)	< 0.001*
Duration of ART initiation, N (%)				< 0.001*
Within 6 months	17075 (36.37)	2086 (82.32)	19161 (38.72)	
> 6 to 12 months	11305 (24.08)	303 (11.96)	11608 (23.46)	
> 12 months	18574 (39.56)	145 (5.72)	18719 (37.83)	
Median (IQR) rapid ART group (months)	0.72 (0.20–2.86)	0.99 (0.33–2.99)	0.72 (0.20–2.86)	< 0.001*
Rapid ART group, N (%)				< 0.001*
Within 1 month	27532 (58.64)	1270 (50.12)	28802 (58.20)	
> 1 to 3 months	7913 (16.85)	632 (24.94)	8545 (17.27)	
> 3 months	11509 (24.51)	632 (24.94)	12141 (24.53)	
Main insurance class, N (%)				< 0.001*
UCS	20005 (42.61)	1223 (48.26)	21228 (42.89)	
SSS	17208 (36.65)	688 (27.15)	17896 (36.16)	
Others	9429 (20.08)	600 (23.68)	10029 (20.27)	
Unknown	312 (0.66)	23 (0.91)	335 (0.68)	
Region, N (%)				< 0.001*
Bangkok	9550 (20.34)	382 (15.07)	9932 (20.07)	
Central	8308 (17.69)	439 (17.32)	8747 (17.67)	
Eastern	5550 (11.82)	331 (13.06)	5881 (11.88)	
Northeastern	11341 (24.15)	640 (25.26)	11981 (24.21)	
Northern	7691 (16.38)	414 (16.34)	8105 (16.38)	
Southern	4514 (9.61)	328 (12.94)	4842 (9.78)	

Abbreviations: AIDS, acquired immunodeficiency syndrome; ART, antiretroviral therapy; CD4, cluster of differentiation 4; DTG, dolutegravir; HIV, human immunodeficiency virus; NNRTI, non-nucleoside reverse transcriptase inhibitor; PI, protease inhibitor; UCS universal coverage scheme; SSS, social security scheme.

Presented as N (%) for categorical data and median (interquartile range) for continuous data.

The comparisons were performed using Pearson's Chi-square tests, for categorical data, and Mann-Whitney U tests for continuous data.

*Significant at 5% level of significance.

Table 2 Distribution of mortality rates among people living with HIV at ART initiation in Thailand by patient characteristics.

Characteristics	Death	person-years	Mortality rate per 100 person-years (95%CI)	<i>p</i>
Overall	2534	39977	6.34 (6.10-6.59)	
Gender				0.458
Male	1808	28879	6.26 (5.98-6.56)	
Female	726	11098	6.54 (6.08-7.04)	
Age group (years)				< 0.001*
15 to 24 years	161	7647	2.11 (1.80-2.46)	
25 to 49 years	1685	26937	6.26 (5.96-6.56)	
≥ 50 years	688	5393	12.76 (11.84-13.75)	
Pre-ART CD4 count (cells/mm³)				< 0.001*
< 200	1796	15925	11.28 (10.77-11.81)	
200 to < 350	190	7773	2.44 (2.12-2.82)	
≥ 350	124	13565	0.91 (0.77-1.09)	
Unknown	424	2714	15.62 (14.21-17.19)	
Viral load after 1 year of ART (copies/ml)				< 0.001*
≤ 50	239	25830	0.93 (0.82-1.05)	
> 50	309	8980	3.08 (3.08-3.85)	
Unknown	1986	5168	36.78 (36.78-40.16)	
First regimen				0.061
NNRTI-based ART	1911	33601	5.69 (5.44-5.95)	
PI-based ART	66	1156	5.71 (4.48-7.26)	
DTG-based ART	453	3877	11.68 (10.66-12.81)	
Others	104	1343	7.74 (6.39-9.39)	
Opportunistic infection at ART initiation				< 0.001*
Yes	434	2547	17.04 (15.51-18.72)	
No	2100	37430	5.61 (5.38-5.86)	
Year of ART initiation				< 0.001*
2021	1614	315949	5.11 (4.87-5.36)	
2022	920	8382	10.99 (10.29-11.71)	
HIV stage				< 0.001*
Asymptomatic HIV	1165	24193	4.82 (4.55-5.10)	
Symptomatic HIV	496	5763	8.61 (7.88-9.40)	
AIDS	873	10022	8.71 (8.15-9.31)	
Duration of ART initiation				< 0.001*
Within 6 months	2086	3835	54.40 (52.11-56.78)	
> 6 to 12 months	303	8654	3.50 (3.13-3.92)	
> 12 months	145	27489	0.53 (0.45-0.62)	
Rapid ART group				< 0.001*
Within 1 month	1270	21938	5.79 (5.48-6.12)	
> 1 to 3 months	632	6839	9.24 (8.55-9.99)	
> 3 months	632	11201	5.64 (5.22-6.10)	
Main insurance class				< 0.001*
UCS	1223	16095	7.60 (7.18-8.04)	
SSS	688	15851	4.34 (4.03-4.68)	
Others	600	7706	7.79 (7.19-8.43)	
Unknown	23	326	7.06 (4.69-10.63)	
Region				< 0.001*
Bangkok	382	8534	4.48 (4.05-4.95)	
Central	439	7226	6.08 (5.53-6.67)	
Eastern	331	4506	7.35 (6.60-8.18)	
Northeastern	640	9367	6.83 (6.32-7.38)	
Northern	414	6641	6.23 (5.66-6.86)	
Southern	328	3703	8.86 (7.95-9.87)	

95%CI-95%confidence interval, ref – reference.

Abbreviations: AIDS, acquired immunodeficiency syndrome; ART, antiretroviral therapy; CD4, cluster of differentiation 4; DTG, dolutegravir; HIV, human immunodeficiency virus; NNRTI, non-nucleoside reverse transcriptase inhibitor; PI, protease inhibitor; UCS universal coverage scheme; SSS, social security scheme.

The comparisons were performed using Log-rank tests, for Kaplan–Meier survival curves.

*Significant at 5% level of significance.

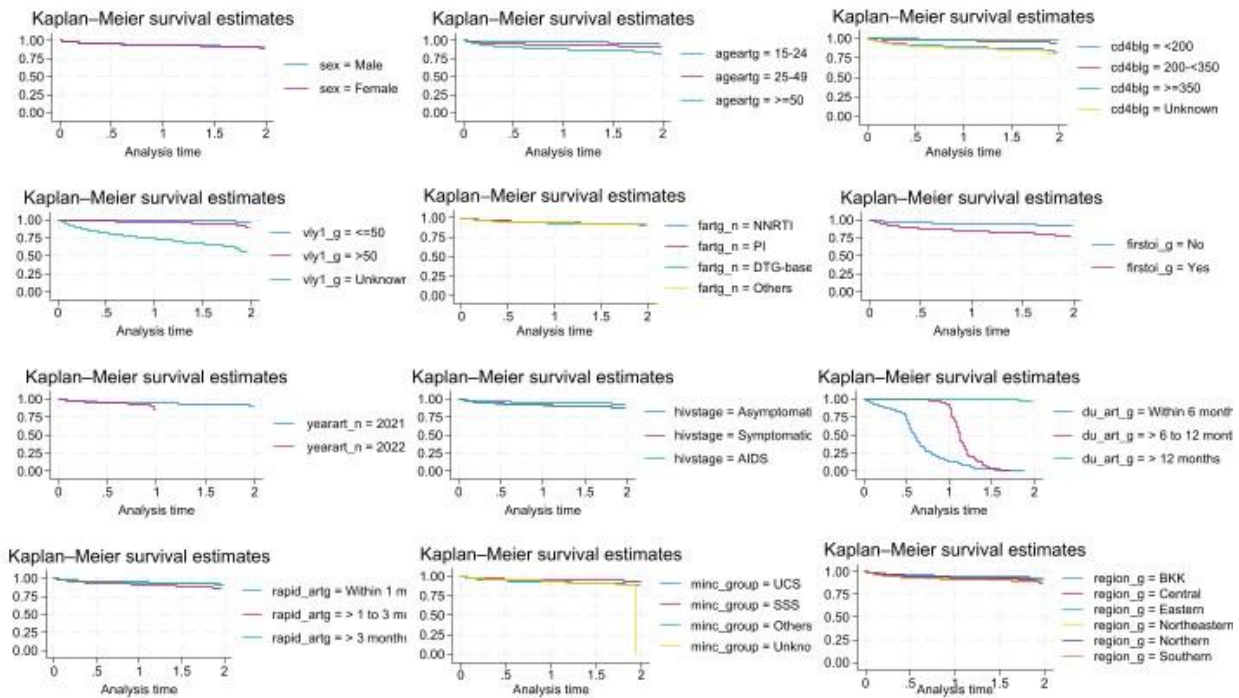


Figure 1. Kaplan–Meier survival curves among people living with HIV at ART initiation in Northern Thailand, by patient characteristics.

Table 3 Assessment of the PH assumption and factors associated with mortality rates among people living with HIV at the initiation of antiretroviral therapy initiation in Thailand.

Characteristics	Multivariable		PH tests	
	aHR (95%CI)	<i>p</i>	χ^2	<i>p</i>
Gender				
Male	ref.		ref.	
Female	0.95 (0.87-1.04)	0.290	4.61	0.032*
Age group (years)				
15 to 24 years	ref.		ref.	
25 to 49 years	2.14 (1.82-2.53)	< 0.001*	8.34	0.004*
≥ 50 years	3.63 (3.04-4.33)	< 0.001*	2.75	0.097
Pre-ART CD4 count (cells/mm³)				
< 200	ref.		ref.	
200 to < 350	0.36 (0.31-0.42)	< 0.001*	18.79	< 0.001*
≥ 350	0.19 (0.16-0.23)	< 0.001*	14.02	< 0.001*
Unknown	0.64 (0.57-0.71)	< 0.001*	5.08	0.024*
Viral load after 1 year of ART (copies/ml)				
≤ 50	ref.		ref.	
> 50	2.07 (1.75-2.46)	< 0.001*	0.40	0.528
Unknown	5.07 (4.34-5.92)	< 0.001*	11.58	0.001*
First regimen				
NNRTI-based ART	2.37 (2.07-2.70)	< 0.001*	5.96	0.015*
PI-based ART	2.56 (1.94-3.37)	< 0.001*	0.20	0.653
DTG-based ART	ref.		ref.	
Others	1.97 (1.59-2.45)	< 0.001*	3.09	0.079*
Opportunistic infection at ART initiation				
Yes	ref.		ref.	
No	0.84 (0.73-0.96)	0.011*	2.06	0.151
Year of ART initiation				
2021	ref.		ref.	
2022	0.37 (0.33-0.41)	< 0.001*	0.70	0.402
HIV stage				
Asymptomatic HIV	0.73 (0.66-0.82)	< 0.001*	4.54	0.033*
Symptomatic HIV	ref.		ref.	
AIDS	0.94 (0.82-1.07)	0.348	< 0.01	0.962
Duration of ART initiation				
Within 6 months	ref.		ref.	
> 6 to 12 months	0.06 (0.05-0.07)	< 0.001*	200.05	< 0.001*
> 12 months	0.002 (0.002-0.003)	< 0.001*	72.82	< 0.001*
Rapid ART group				
Within 1 month	ref.		ref.	
> 1 to 3 months	1.05 (0.96-1.16)	0.284	3.35	0.067
> 3 months	1.03 (0.94-1.14)	0.437	7.96	0.005*
Main insurance class				
UCS	1.02 (0.92-1.16)	0.748	17.93	< 0.001*
SSS	ref.		ref.	
Others	0.82 (0.73-0.93)	0.002*	2.95	0.086
Unknown	1.12 (0.73-1.68)	0.633	9.26	0.002*
Region				
Bangkok	ref.		ref.	
Central	0.86 (0.75-0.99)	0.037*	0.17	0.678
Eastern	0.92 (0.79-1.07)	0.277	0.08	0.774
Northeastern	0.91 (0.80-1.04)	0.184	3.33	0.068
Northern	0.98 (0.85-1.13)	0.762	5.47	0.019*
Southern	1.07 (0.92-1.25)	0.382	0.39	0.535

95%CI-95%confidence interval, ref – reference.

Abbreviations: AIDS, acquired immunodeficiency syndrome; ART, antiretroviral therapy; CD4, cluster of differentiation 4; DTG, dolutegravir; HIV, human immunodeficiency virus; NNRTI, non-nucleoside reverse transcriptase inhibitor; aHR, adjusted Hazard ratio; PI, protease inhibitor; UCS universal coverage scheme; SSS, social security scheme.

*Significant at 5% level of significance.

Table 4 Stratified Cox PH model of factors associated with mortality rates among people living with HIV at ART initiation in Thailand.

Characteristics	Multivariable	
	aSCHR (95%CI)	<i>p</i>
Gender		
Male	ref.	
Female	0.95 (0.87-1.04)	0.255
Age group (years)		
15 to 24 years	ref.	
25 to 49 years	2.18 (1.84-2.58)	< 0.001*
≥ 50 years	3.43 (2.86-4.12)	< 0.001*
Viral load after 1 year of ART, N (%)		
≤ 50	ref.	
> 50	2.21 (1.85-2.65)	< 0.001*
Unknown	6.02 (5.12-7.09)	< 0.001*
First regimen		
NNRTI-based ART	2.53 (2.21-2.90)	< 0.001*
PI-based ART	2.51 (1.88-3.36)	< 0.001*
DTG-based ART	ref.	
Others	2.07 (1.65-2.58)	< 0.001*
Opportunistic infection at ART initiation		
Yes	ref.	
No	0.85 (0.74-0.98)	0.028*
Year of ART initiation		
2021	ref.	
2022	0.37 (0.33-0.41)	< 0.001*
HIV stage		
Asymptomatic HIV	0.81 (0.73-0.91)	< 0.001*
Symptomatic HIV	ref.	
AIDS	0.97 (0.84-1.11)	0.658
Region		
Bangkok	ref.	
Central	0.91 (0.79-1.05)	0.197
Eastern	0.92 (0.79-1.08)	0.301
Northeastern	0.95 (0.83-1.09)	0.477
Northern	1.03 (0.89-1.19)	0.687
Southern	1.08 (0.92-1.25)	0.362

95%CI-95%confidence interval, ref – reference.

Abbreviations: AIDS, acquired immunodeficiency syndrome; ART, antiretroviral therapy; CD4, cluster of differentiation 4; DTG, dolutegravir; HIV, human immunodeficiency virus; NNRTI, non-nucleoside reverse transcriptase inhibitor; aHR, adjusted Hazard ratio; PI, protease inhibitor; UCS universal coverage scheme; SSS, social security scheme.

*Significant at 5% level of significance.

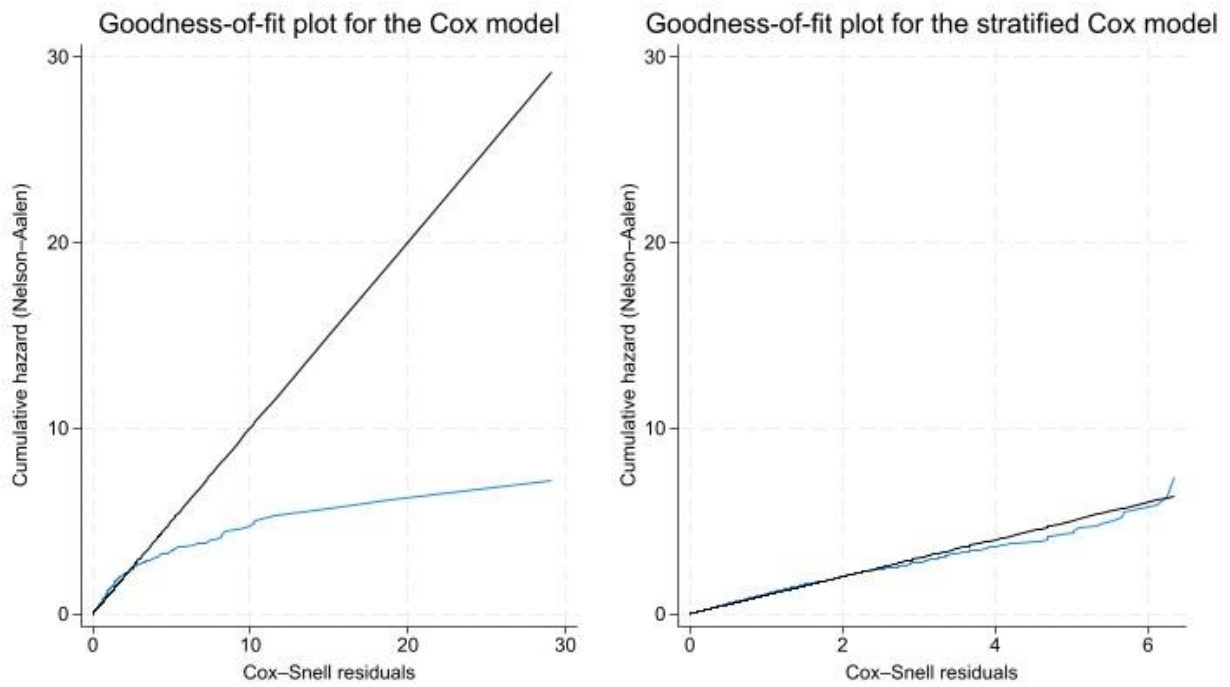


Figure 2. Goodness-of-fit plot for Cox regression and Stratified Cox PH model.

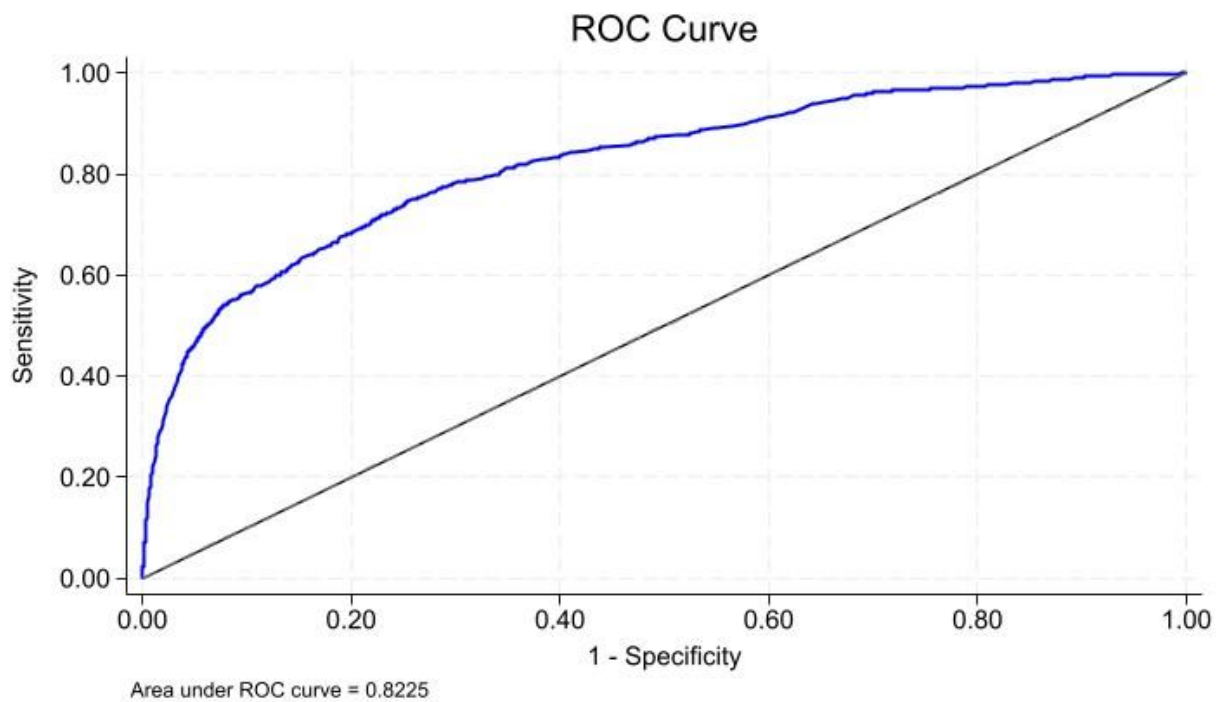


Figure 3. ROC curve of Stratified Cox PH model.

STAT/MP 18.0: Stratified Cox PH model.

Code:	
Table 1 Characteristics of people living with HIV at ART initiation in Northern Thailand by mortality status.	
Characteristics table	<pre> dtable, by(died, tests testnotes total) sample(, statistic(frequency percent) place(seplabels)) continuous(ageart cd4bl log_vly1 du_art_m rapid_art_m, statistics(median p25 p75) test(kwallis)) factor(sex ageartg cd4blg vly1_g fartg_n firstoi_g yearart_n hivstage du_art_g rapid_artg minc_group region_g, statistics(fvfrequency fvpercent) test(pearson)) nformat(%9.2f median p25 p75 fvpercent) sformat("%s") fvpercent p25 p75) </pre>
Table 2 Distribution of mortality rates among people living with HIV at ART initiation in Northern Thailand, by patient characteristics.	
Deaths	<pre> tab sex died if died == 1, col tab ageartg died if died == 1, col tab cd4blg died if died == 1, col tab vly1_g died if died == 1, col tab fartg_n died if died == 1, col tab firstoi_g died if died == 1, col tab yearart_n died if died == 1, col tab hivstage died if died == 1, col tab du_art_g died if died == 1, col tab rapid_artg died if died == 1, col tab minc_group died if died == 1, col tab region_g died if died == 1, col </pre>
person-years & Mortality rate per 100 person-years (95%CI)	<pre> strate , per(100) strate sex, per(100) strate ageartg, per(100) strate cd4blg, per(100) strate vly1_g, per(100) strate fartg_n, per(100) strate firstoi_g, per(100) strate yearart_n, per(100) strate hivstage, per(100) strate du_art_g, per(100) strate rapid_artg, per(100) strate minc_group, per(100) strate region_g, per(100) </pre>
Log-rank test	<pre> sts test sex sts test ageartg sts test cd4blg sts test vly1_g sts test fartg_n sts test firstoi_g sts test yearart_n sts test hivstage sts test du_art_g sts test rapid_artg sts test minc_group sts test region_g </pre>
Table 3 Assessment of the proportional hazards assumption and factors associated with mortality rates among people living with HIV at the initiation of antiretroviral therapy initiation in Thailand.	
Univariable	<pre> stcox i.sex stcox i.ageartg stcox i.cd4blg stcox i.vly1_g stcox ib3.fartg_n stcox i.firstoi_g stcox i.yearart_n stcox ib2.hivstage stcox i.du_art_g stcox i.rapid_artg stcox ib1.minc_group stcox i.region_g </pre>
Multivariable	<pre> stcox i.sex i.ageartg i.cd4blg ib3.fartg_n ib1.firstoi_g i.yearart_n b2.hivstage i.du_art_g i.rapid_artg ib1.minc_group i.region_g, baselevel </pre>
Propotional Hazards assumption	estat phtest, detail

STAT/MP 18.0: Stratified Cox PH model. (continued)

Code:	
Table 4 Stratified Cox proportional hazards model of factors associated with mortality rates among people living with HIV at ART initiation Thailand.	
Multivariate Stratified Cox proportional hazards model	stcox i.sex i.vly1_g ib3.fartg_n ib1.firstoi_g i.yearart_n i.ageartg ib2.hivstage i.region_g, strata(cd4blg du_art_g rapid_artg minc_group) baselevel
Graphs	
Kaplan-Meier survival graph	sts graph sts graph, by(sex) sts graph, by(ageartg) sts graph, by(cd4blg) sts graph, by(vly1_g) sts graph, by(fartg_n) sts graph, by(firstoi_g) sts graph, by(yearart_n) sts graph, by(hivstage) sts graph, by(du_art_g) sts graph, by(rapid_artg) sts graph, by(minc_group) sts graph, by(region_g)
Goodness-of-fit plot	estat gofplot
ROC curve	predict xb, xb roctab _d xb